SJF PRIMITIVE

```c
#include <stdio.h>

int main()
{
    int n,i,time=0,completed=0,smallest;
    int at[10],bt[10],rt[10],wt[10],tat[10],p[10];
    float avgwt=0,avgtat=0;

    printf("Enter number of processes: ");
    scanf("%d",&n);

    for(i=0;i<n;i++)
    {
        p[i]=i+1;
        printf("Enter Arrival Time and Burst Time for P%d: ",p[i]);
        scanf("%d%d",&at[i],&bt[i]);
        rt[i]=bt[i];
    }

    while(completed<n)
    {
        smallest=-1;

        for(i=0;i<n;i++)
        {
            if(at[i]<=time && rt[i]>0)
            {
                if(smallest==-1 || rt[i]<rt[smallest])
                    smallest=i;
            }
        }

        if(smallest==-1)
        {
            time++;
        }
```

```c
        else
        {
            rt[smallest]--;
            time++;

            if(rt[smallest]==0)
            {
                completed++;

                tat[smallest]=time-at[smallest];
                wt[smallest]=tat[smallest]-bt[smallest];

                avgwt+=wt[smallest];
                avgtat+=tat[smallest];
            }
        }
    }

    printf("\nP\tAT\tBT\tWT\tTAT\n");

    for(i=0;i<n;i++)
    {
        printf("P%d\t%d\t%d\t%d\t%d\n",p[i],at[i],bt[i],wt[i],tat[i]);
    }

    printf("\nAverage Waiting Time = %.2f",avgwt/n);
    printf("\nAverage Turnaround Time = %.2f",avgtat/n);

    return 0;
}
```

"C:\Users\BMSCECSE-L3-56\D

```
Enter number of processes : 4
Enter Arrival Time and Burst Time for P1: 12
10
Enter Arrival Time and Burst Time for P2: 11
13
Enter Arrival Time and Burst Time for P3: 14
15
Enter Arrival Time and Burst Time for P4: 09
15

P       AT      BT      CT      TAT     WT
P1      12      10      22      10      0
P2      11      13      34      23      10
P3      14      15      62      48      33
P4      9       15      47      38      23

Average Waiting Time = 66.00
Average Turnaround Time = 119.00

Process returned 0 (0x0)   execution time : 15.396 s
Press any key to continue.
```